

ActInf Livestream #014.0 ~ The Math is not the Territory: Navigating the Fr

Livestream | The Math is not the Territory: Navigating the Free Energy Principle | 1:46:56

hello

everyone welcome to the active inference

lab the active inference live stream

this is

active inference live stream number 14.0

and i'm looking forward to today's

discussion

welcome to the active inference lab

everyone

we are an experiment in online team

communication

learning and practice related to active

inference

you can find us on our website

on twitter our email address on our

youtube channel

our key base public team and our shared

username active inference

this is a recorded and an archived live

stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome here

and as far as video etiquette for live

streams please mute if there's noise in

the background

raise your hands so we can hear from

everybody who wants to speak

and we will use respectful speech

behavior

for the regularly scheduled active

inference streams during 2021

all group discussions are going to be on

tuesdays

from 7 to 9 00 a.m pacific time and that

will be a regular weekly time slot
and we also do special sessions if you
go to this
link with the red text you can see some
information about participating in
future streams and definitely get in
touch with us
if you want to participate in 2021 or if
you have an idea for an event
or somebody we could invite for a live
stream
if you go to that spreadsheet you will
see these
um columns that cover the date of each
active inference live stream that's
scheduled and what we'll be reading and
anything else about it we just finished
up
13.1 and 13.2 which were really fun and
this dot zero video is heading into
active infrared stream 14.1 and 14.2
reading the math is not the territory by
mel andrews
and mel will be joining us for these
discussions so we're really
looking forward to that today
in active inference stream 14.0 the goal
is to set
the context for 14.1 and 14.2 which will
be group discussions
and the paper under consideration in all
three of these live streams
is the math is not the territory
navigating the free energy principle
by mel andrews 20 20. the video
is just an introduction to some of the
ideas and some of the context it's not a
review

or a final word and a punchline
for this paper is all models are wrong
but some are useful and some are the
free energy principle
which ones are which that's what we're
here to talk about
in the sections of 14.0 today
first we'll go through the aims and
claims of the paper
talk about a little background
represented by the keywords
go through the abstract and look over
the roadmap
then we'll pass through a few key topics
and key quotations from the paper
and end up with a couple questions that
are remaining
in 14.1 and in 14.2 we're going to be
discussing this very same paper
so save your questions post them as a
comment on this video
and definitely get in touch if you want
to participate live
in this conversation or in a future or
different conversation
okay here we are in the paper which is
the math is not the territory
navigating the free energy principle by
mel andrews
which was put up on a philsai archive
pre-print server on october 8th 2020
and the aims and the claims of the paper
are as follows
quoting mel with respect to the
demarcation problem
i defend the position that the fep free
energy principle
can be taken to belong properly to the

domain of science
and we're going to talk more about the
demarcation problem in a few minutes but
that's the question of what is science
and what is not science
mel writes the fep is certainly not
anything like a theory
a law a hypothesis or a research program
as these have been classically
understood in the history and philosophy
of science
so this is a claim that what the fep is
or what it does how it's enacted in the
world is different than some of these
classical
history and philosophy of science terms
like theory law hypothesis
or program in order to see how the
literature on scientific models this is
a claim
accords with the fep however we will
first need to establish
a firm sense of what the fep is and what
it is not
this will require dispelling a number of
false assumptions that have been made
about the framework
i accomplished this first by tracing out
the historical buildup to the fep
illustrating where the formalism has
been derived from and from what it has
come to signify
and continuing on that quote this serves
the purpose primarily of showing how
much of the mechanics of the fep
had physical meanings in its initial
form but has since come into a strictly
formal use

forma following this i will demonstrate
that the formalism is empty of any sort
of facts or assertions
about the state of nature that would
allow it to draw taxonomical
distinctions
to differentiate classes of natural
systems to explain their behavior in
terms of underlying mechanisms or
to bring forth testable hypotheses
strong
claims and that's what's going to be fun
about this paper
and this discussion is it's uh hopefully
going to be
highlighting many different perspectives
on what the fep
is or is not so if you're watching live
if you're watching afterwards if you
want to participate in some way
and you're curious about these questions
these are the kinds of questions that
are being actively
researched and debated so you're in the
right place
let's see how we're going to set up the
background
to address these aims and claims of the
paper
the keywords of the paper are as follows
free energy principle
bayesian brain life mind continuity
scientific models modeling and
simulations scientific theories
epistemic virtues falsification and the
demarcation problem
now as always for these keywords there's
really so much to be said about each one

you could study for your whole life
going down just one detail of one of
these keywords so it's not a review
it's not a final word my target with
these keyword summaries
is the construction of a bi-directional
on-ramp
it's a it's not an off-ramp it's
actually an on-ramp for everyone
and the on-ramp is going between this
paper and broader communities
so if you're in the active inference
community i hope that this paper
is an introduction or a reminder of some
of the bigger
philosophical and historical trends that
the fep
and active inference research is
happening inside of
and if you're outside of the active
inference research community
then hopefully you're going to find a
few points of contact with how you
currently view the world
but also start to get excited and
curious about this little thing
called free energy principle what is the
free energy principle
well on this slide i put up a couple of
terms
and these terms are all going to come
together in the free energy principle
and i'll read a couple quotes on that
but the terms up here
are multi-scale systems from the left to
the right so that means
systems that exist across multiple
spatial and temporal scales

so something that exists over
milliseconds and seconds and minutes
that's temporal scales and something
that is exist
existing or displaying orderliness over
spatial scales that are small to big so
cells that are part of a
organized tissue that are part of an
organism part of a
ecosystem that's multi-scale
another key aspect of the free energy
principle is a focus
on ensembles which is akin to talking
about collective behavior
so we're not going to be behavioral
reductionists or
essentialists we're not going to go to
the smallest possible unit of motion
and say that that's the most meaningful
nor are we going to simply go to the
largest
scale of phenotype and say that that's
the most meaningful
we're going to be using this multi-scale
perspective
and thinking about ensembles of
collective systems
and using that to move beyond
internalism and externalism
to integrate holism and reductionism
we're also going to be taking a strong
informational thermodynamic perspective
and some of the history of this we're
going to trace in this video and was so
excellently traced by mel
during the paper and info thermodynamics
is basically
starting with thermodynamics so about

the vibrations of molecules and about
heat and entropy
and then considering how since
information is something that has to be
embodied or manifest in the real world
for example on a hard drive or passing
through a wire
we can actually think about
thermodynamics and signal processing
within a broader informational capacity
and then this last point here action
policy selection
as so this first part is saying it's
going to be a control theory or a
cybernetic
focused framework and the way that those
policies are going to be obtained
is going to be through a type of
variational multi-scale bayesian
inference so we're not going to be using
uh linear regression or we're not going
to be using
some other type of machine learning
algorithm we're going to be using
derivatives on this variational bayesian
inference approach
how about this bug on the left side of
this slide
why is there a little bug well that
comes from
the title of an interview that i carried
out with my colleague martin
and carl firsten in 2018 that was
entitled of woodlice and men
a bayesian account of cognition life and
consciousness at the url
here and we asked him early in the
interview we said

what is the free energy principle and
how did you
get to the place where you are at with
it now
and carl wrote he was watching
wood lice which are these little bugs
watching them scurry around
on the ground and he wrote after half an
hour of observation and innocent
child-like contemplation i realized that
their scurrying had no purpose
or intent they were simply moving faster
in the sun and slower in the shade
which meant that they were spending most
of their time in the shade from the
population or ensemble
perspective because they would go
rapidly through the sun and then dwell
in the shade so even if it was 50 50
you'd end up enriching them in the shade
the simplicity of this explanation for
what one could artfully call
biotic self-organization appeal to me
then and appeals to me now
it is exactly the same principle that
underwrites the ensemble density
dynamics of the free energy principle
and all of its corollaries so that's a
key piece
there's going to be ensemble density
dynamics so it's going to change through
time
and it's going to be collective behavior
and it's going to be
related to not just the free energy
principle but some of its corollaries or
some things that are
adjacent to it and we'll talk about what

those are

he then continued over the subsequent 20

years i learned enough mathematics to

think about the shapes

of wood lice in terms of density

dynamics

namely the evolution of probability

density distributions over ensembles of

states

for example swarms of wood lice happily

people had been using this

exact sort of framework both to model

the world and analyze their data

i came to know this as ensemble learning

and in particular

variational base this is how the free

energy principle

developed into the current framework in

brief i was very lucky to meet the right

people

and work in an era when these ideas were

in the air

so as always humble and insightful

carl what is the bayesian brain

let's draw from one of the questions

that martin and i

asked in our interview because we tried

to be very clear about what the bayesian

brain was

so that we could be very clear in

distinguishing it what the free energy

principle was

and also just so that we don't end up

confounding our uncertainty

as we try to contrast different ideas

because it's really important to have a

solid understanding

of these terms that we're going to be

talking about so that we don't accidentally make errors or misattributions or false positives or false negatives in our understanding

we wrote the bayesian brain hypothesis and the free energy principle on the basis of predictive coding and the free energy principle are often equated with one another

you have yourself suggested that these three frameworks are variations of the same basic mechanisms to be clear what we call the bayesian brain hypothesis is the idea that the brain performs inference according to bayes theorem

integrating new information in light of existing models of the world

a perceptual or cognitive state can be modeled as being a posterior probability so $p(h \text{ given } d)$ where p stands for probability h hypothesis cause

and d observed or available data so the probability of the hypothesis being the case given the data

the posterior probability is the product of the likelihood $p(d \text{ given } h)$ the probability of the data given the hypothesis

and the prior probability which is $p(h)$ or just the probability of a hypothesis being true

in other words the probability of the model h being true is the likelihood of the model h given

the observation d
multiplied by the likelihood of a model
 h relative to other models under
consideration
to make these equations a bit more
concrete let us take the following
example the brain is receiving scarce
data
 d from the retina and must form a model
 h of how the world is has caused this
pattern on the retina
in bayesian terms the problem to be
solved is the following
 $p(h \text{ given } d)$ or $p(h|d)$
of h given d the bayesian brain
hypothesis
implies that the posterior probability
at time one or t_1
provides the prior probability at time
 t_2
so this was our characterization of the
bayesian brain
and we're going through it because it's
a key word in this paper because it's
very
important to understand what exactly the
bayesian brain is
or isn't so we can understand what the
fep is or isn't
so that's the bayesian brain hypothesis
it's the idea that we're going to use
bayesian approaches bayesian equations
formalisms
to model the way that the brain as a
statistical instrument
in the world yes an evolved one but
we're going to focus on the process of
inference for now

it's going to be generating hypotheses
about latent causes of the world or
about things in the world
given observations and updating its
prior understandings or beliefs about
the world

we then asked if carl agreed with that
characterization of the bayesian brain
hypothesis

and he wrote even though he also
improved the question

as we had asked it he wrote in his
response do i agree

with this characterization of the
bayesian brain hypothesis yes i do
with a couple caveats i think it is
useful to make a fundamental distinction
at this point that we can appeal to
later

the distinction is between a state and a
process theory

i.e the difference between a normative
principle that things may or may not
conform to

and a process theory or hypothesis about
how that principle is realized
under this distinction the free energy
principle ergo

it's a normative principle stands in
stark distinction to things like
predictive coding and the bayesian brain
hypothesis so critical point principles
are contrasted with hypotheses
this is because the free energy
principle is what it is a principle
like hamilton's principle of stationary
action it cannot be falsified

key point and this has resulted in a lot

of discussion

it cannot be disproven in fact there's
not much you can do with a free energy
principle

unless you ask whether measurable
systems conform to the principle
on the other hand hypotheses that the
brain performs some type of bayesian
inference or predictive coding
are what they are hypotheses these
hypotheses

may or may not be supported by empirical
evidence so in other words

those hypotheses can be falsifiable they
can be tested and rejected

you can say if it's the case that the
brain is doing bayesian inference it
should have this

observation in this brain region okay we
measured it's consistent or it's not
we rejected it so it's funny because
even though we're talking
about bayesian brain which is a
post-falsificationist idea

in the terms in the sense of statistics
we're actually going to use
the popperian idea of falsification from
the philosophy of science

perspective to ask whether we could
falsify the bayesian brain idea

so what does it look like to have
hypotheses that are beyond
falsificationism

does bayesian statistics or the bayesian
turn actually

move us away from simple falsification
because it's not the case that there's
just going to be some experiment in most

situations that slam dunk disproves
usually most evidence is actually
ambiguous
depending on how you weight the
different possibilities
for example a tree to an evolutionary
biologist
is an example of natural selection but
to a creationist the very same tree is
an example of creation
so that's how evidence is related to the
generative model that someone has
in this view the relationship between
the free energy principle
and predictive coding or the bayesian
brain or active inference
is the relationship between a principle
and a process theory
crucially there are lots of process
theories that conform to the free energy
principle
predictive coding is arguably the
predominant process theory in cognitive
neuroscience
however there are other contenders for
example based upon discrete as opposed
to continuous state space models
these would include things like belief
propagation and variational message
passing
these schemes or processes serve as
plausible metaphors
for neuronal message passing that may or
may not have the look and feel of
predictive coding
it is important to note that there have
been other process theories that have
not fared so well in light of empirical

evidence

for example probabilistic population

codes and attempts to understand

ensemble dynamics in terms of sampling

from the posterior

e.g gibbs sampling and particle

filtering so in other words

free energy principle is beyond or

before

empirical data however it lends itself

to

corollary hypotheses process theories

that make specific

testable predictions about how the fep

is getting done

and so you could predict that little

tiny unicorns are doing it and then you

can go and measure and falsify that idea

so again we're using the paparian idea

of falsification

of theories to discuss whether this

bayesian

perspective on the brain is adequate

the bayesian understanding of the brain

if we pull back a step

is related to what is called and what is

keyworded as

life mind continuity and this is what

the definition of the inactive life mind

continuity thesis

is it's written as any living system

is ipso facto a cognitive system or in

other words

life is sufficient for mind so that's

the claim that living systems are

mindful systems what makes a system

living and therefore cognitive

is its autonomy now this often

ties me up because i think okay
autonomous systems does that mean they
must be cognitive so now we're working
backwards from what appears to be
cognitive
but isn't that hubris because we're
putting our understanding of another
system
in front of its actual way of being in
the world i don't know
and the last point here is understanding
cognition requires
understanding the principles of
biological atta
autonomy and this uh is
just the simplest phrasing of that
thesis so that's kind of like
that claim which is living systems are
those with mind
now where does awareness come into play
free will agency
autonomy great questions if anyone knows
if anyone has suggestions let's
hear them in the live chat or come on
and let's discuss it
on a stream um but this was something
that i was just
learning about when i was preparing for
this talk
and another thing that i found that was
pretty cool was a paper from 2017
in progress in biophysics and molecular
biology which sounds like it would be
just about molecular biology
of a journal but actually it's an
awesome journal with a lot
other ideas happening and it's a paper
by

um uh hippolyto and martins
and ines hippolyto is also a free energy
principle researcher so i thought it'd
be cool to
highlight this paper and this
perspective because
the inactive life-mind continuity thesis
is not
written about with the fep in mind
whereas this paper may have been written
at least
a little bit more directly with fep in
mind and it's called
mind life continuity a qualitative study
of conscious experience
and the authors wrote according to the
mind life continuity thesis there are
three levels
on the weak continuity view whatever has
mind
will have life although not all things
that have life have mind
so this is the weakest version of the
claim which is it isn't the case that
just being life
makes you mindful however if you have
mind you do have life so that's the
weakest version of the claim
on the strong continuity view life and
mind have a common abstract model or set
of basic organizational properties or
things that
lead to them so in this view the
one-to-one overlap
is strong there's a total venn diagram
overlap living systems are mindful
systems and that's one-to-one
so you're either both or you're neither

on the methodology continuity view the understanding of mind requires further understanding the role that it plays within the whole living system

so that's a little bit more like a meta or a holist or an ecological perspective because it's like saying we need to understand autonomy and agency which is the embeddedness of the agent in the niche we can't just look at the head and ask whether mind is inside of it

it's actually some a view that suggests that life and mind are relational so that's pretty cool

each of these views within the life-mind continuity thesis share the perspective that cognition should be studied in the context of the whole organism

so an activism holism ecological psychology

a lot of these terms are coming into play with this life-mind continuity idea let's talk about scientific models so scientific models it's a pretty big field there's a lot of different types of models and science means a lot of different things we're going to talk more about what science is or isn't in a couple of minutes but it's times like these

when we can just go to the simple english wikipedia page and try to find the truth and on simple english wiki

it was written a scientific model is a
simplified abstract view of a complex
reality

go to reality keyword if you want to
find out what reality is i guess

scientific models are used as a basis
for scientific work

they can be used to explain predict and
test

or to develop computer programs or
mathematical equations

or as i always think about it model
explain predict design

perturb control a scientific model
represents complex

objects events and physical processes in
a logical way

they are an image of an original which
can be a model itself

so this is kind of like simulacra theory
uh or the work of umberto echo

scientific models

only have those details of the object or
image

modeled that are relevant i'd take a
slight contention there

it's not that they are only capturing
the details of the object or the image
that are relevant it's that they capture
what they capture

and whether what is captured is relevant
or irrelevant that's perspectival

or whether it's useful or not it's also
up to the

um the eye of the model viewer let's say
and there is no strict mapping between a
model

and the original object it models models

may only be valid for a given time

interval for a given object

or a given purpose for example there

isn't necessarily going to be the one

true

bitcoin price model there might be

different time scales of analysis

or different kinds of trading systems

that lend themselves to different kinds

of models being utilized

and a model trained on year one through

five might

might not be as good or might not even

be adequate for a different time period

so models are situational and that's

also

where we get this idea of science as

something that's doing relevant and

useful work utilitarian pragmatic work

yet also is a social construct because

it's enacted in a social setting

here's a couple um flow charts that are

just showing different aspects of the

scientific

process and the role that models play in

the scientific process

basically starting from a real system

models are made

and experiments can be carried out

models lead to

models of the system now model system

gets used a little bit confusingly for

example to refer to fruit flies as a

model system

as in their system that people study but

fruit flies are a real

system and a model system would be like

a simulation of a fruit fly

but of course you get into these
questions like what if i'm really
actually interested in studying the
model of a fruit fly
but whether you're doing experiments in
a so-called real system
or simulations in a so-called model
system
you're going to end up with observations
and then even more abstract than
simulation is actually
doing analytical work or um
or theory above the level of the model
system
to obtain theoretical predictions like
for all temperatures
this variable is going to be greater
than this variable and then
by comparing and contrasting the
empirical experimental results
the simulations and results that we
observe and theoretical predictions
that we might get from a more a priori
source
we're able to compare and improve our
model and theory overall
so this is not quite the scientific
process that's on the wall of the
elementary school
but it does capture how scientific
modeling is about
iterated improvements and about
integrating across different kinds of
experiments
different kinds of simulations and just
to give a point on the falsification
because falsification means in some
situations to

um to confabulate
or maybe fabulate to to make up to um
generate data that's false like to
basically commit scientific fraud is
often called falsification of data
so if someone says those documents were
falsified they don't mean that it was a
hypothesis that got tested
usually they mean somebody deep faked or
disenforced those documents
but here is falsifiability from this
paparian perspective
in the philosophy of science
falsifiability false
falsifiability or refutability is the
capacity for a statement theory or
hypothesis to be contradicted by
evidence
for example the statement all swans are
white
is falsifiable because one could observe
that black swans exist
falsifiability was introduced by the
philosopher of science karl popper
in his book logic der forschung 1934
revised and translated into english in
1959 as the logic of scientific
discovery
he proposed it as the cornerstone this
idea of falsification
to both the problem of induction and the
problem of demarcation
so induction is about generalizing from
the specifics to the generals
but what is this problem of demarcation
we knew it's one of the key words
so what is it another wikipedia coming
in to save the day

for these last minute presents and
presentation makers like myself
the demarcation problem in the
philosophy of science and epistemology
which is the philosophical field about
knowledge stem
knowledge not just science technology
engineering
and mathematics but stem means knowledge
and epi
means above so the philosophical field
that's above
the question of knowledge in the
philosophy of science in epistemology
the demarcation problem is the question
of how to distinguish between science
and non-science
it is it examines the lines between
science pseudoscience
and other products of human activity
like art literature and beliefs
the debate continues after over two
millennia of dialogue
among philosophers of science and
scientists in various fields
the debate has consequences for what can
be called scientific in fields such as
education
and public policy so at the social level
this is why it's important
because it will often be the case that
somebody will say well we need a science
driven policy or a science informed
policy or that's unscientific policy
related to public health or education
and so who exactly gets to speak for
science
and who gets to demarcate or delineate

or define

science and who gets to say whether
science is all good and if people mess
up then they're not doing it right or is
science all bad and all the good stuff
comes from some other

question of activity that humans do so
it's actually quite

a big issue about this demarcation
problem

and that is resonant with the fact that
people have been debating for a long
long time and not come to any
simple distinctions now

with this paper in mind we can look at
the first lines of the introduction of
the paper

which is a bold start by mel gotta love
it with respect to the demarcation
problem

i defend the position that the free
energy principle

can be taken to belong properly to the
domain of science

so that's big and unique claim one which
is

fep is within the domain of science now
it might not be

exactly like some of these things that
we've seen before but

it's going to be taken to be within the
domain of science

so if you're listening to this and you
think well i disagree i don't think that
fep is within the realm of science or
maybe you have a yes and

well yes i think it's in the domain of
science and i think it's also in the

domain of something else
great so if you have a yes and or a
disagreement or a question
put it in the live chat or come onto
these group discussions and talk it out
because mel will probably be really
happy to hear about it
and we'll all learn so much by bringing
out these different perspectives on
science because we're
going to some of the fundamental
questions in philosophy and in science
with these papers
let's talk about one way that this
demarcation
problem has been resolved or addressed
in the past
and that's this noma idea or
non-overlapping
magisteria which is the name of a paper
by stephen j
gould who is a evolutionary biologist i
guess we could call him
and gould wrote the net or magisterium
it's kind of like
the the reign of science covers the
empirical realm
what is the universe made of fact and
why does it work that way
theory the magisterium of religion
extends over questions of ultimate
ultimate meaning and moral value these
two magisteria do not overlap
nor do they encompass all inquiry
consider for example the magisterium of
art and the meaning of beauty
now as always when quoting somebody or
reading someone's paper we can

really respect the author's view their
perspective they set it out there how
they wanted it to be said
stephen jay gould chose his words
carefully when he wrote this
but also we can ask whether we agree or
disagree
so here's this non-overlapping
magisteria as a venn diagram
they're so non-overlapping that they're
just separate
circles so on the science side we get
what is the universe made of that's what
that's fact in
gould's mind and why it works that way
which is how or theory or potentially
a scientist might call that mechanism
and then under the noma model
religion is what deals with morality and
metaphysics
now interestingly when some people think
about non-overlapping magisteria
they'll say something like well science
says um
what the world is and and um implicitly
they mean how as well like what the
world is it's also mechanism and
religion says why
and so that's a what how versus why
distinction
for the demarcation so these are
questions that
everyone can think of whether you're
just learning about these topics or
whether you've been
practicing in one field or another for a
long time
it's an interesting thing to wonder

about what is science
what isn't science who decides and why
does it matter
is science one-to-one with technology or
are there technologies that aren't
science or are those two
bound together more often than not and
when we really explore
these maps of meaning that we have for
these different fields
we start to get a better perspective on
oh yeah so when somebody says that it's
a science-informed policy
they want me to think it's good right
they they don't they don't want me to
think that it's
faith-based policy because faith-based
then
dot dot dot dot so it's always a
cultural context
that these terms are understood within
and it's sort of like a battlefield of
meaning
for who gets to control these different
terms but anyways i just
really like that mel led so clearly and
strongly
as a philosopher of science who's so
embedded within our
communities as well and it just
it it uh rides the line with scientific
understanding and philosophical depth
so speaking of philosophical depth and
scientific understanding
let's talk about epistemic virtues and
yet another
wiki um selection i believe it's our
last one

being an epistemically virtuous person
is often equated with being a critical
thinker
and focuses on the human agent and the
kind of practices that make it possible
to arrive
at the best accessible approximation of
the truth
so i wasn't too familiar with this term
epistemic virtues
but it's a little bit like saying what
are the
good attributes or the adjectives that
are good
ver you know not the ones that aren't
good the ones that are
good are the ones that are the epistemic
virtues
virtues and so a couple of them are
written down conscientiousness which is
funny in light of our discussion on the
big five
personality traits in um active 13
as well as a couple of these other ones
that you see here such as
creativity curiosity humility
people can go into debates about what
each of these terms mean or which ones
are or aren't virtues
but the question i would leave us with
at least in this introduction
section is how do we facilitate
epistemic virtue so
given our own understanding how do we
not just develop it
in ourselves but make communities where
epistemic virtue can thrive in an
accessible

fashion all right let's get to the
abstract of the paper
mel wrote the free energy principle or
fep has seen extensive philosophical
engagement
both from a general philosophy of
science perspective and from the
perspective of philosophies of specific
sciences
cognitive science neuroscience and
biology so
people are talking about it the
literature on the fep has attempted to
draw out specific
philosophical commitments and
entailments of the framework
so what are the commitments that's
leading up to
this the ceremony right what's the
commitment that you make leading up to
the ceremony
and then the entailment what follows
from the ceremony
so what leads one to be committed to fep
and what is entailed by committing to
the fep
but the most fundamental questions from
the perspective of philosophy of science
in general
remain open so this is a list of awesome
questions
so each one of these questions we could
probably have a whole group conversation
on
and they're such wonderful questions so
i really
like how this paper was phrased to what
disciplines

does the fep belong does it make
falsifiable claims
what sort of scientific object is it is
it to be taken as a representation of
contingent states of affairs in nature
does it constitute knowledge what role
is it intended to play in relation to
empirical research
and does the fep even properly belong to
the domain of science
but it seems like from the introduction
that mel's answer is yes
i'm uh maybe going to put it in my five
cents at the end of this talk
but it seems like it's good if it's in
science but i'm not too attached
abstract to the extent that it has
engaged with these questions
at all the extant literature has begged
dodged
dismissed and skirted around these
questions without ever
addressing them head on look at these
spatial metaphors
and social metaphors begging dodging
skirting
head-on just wonderful these questions
must i urge be answered satisfactorily
before we can make
any headway on the philosophical
consequences of the fep
so in other words if we don't understand
how we got here and where we're at
we won't be able to make headway as we
move forward with the fep
so if we don't know the past and if we
don't know the present we're not going
to be able to design

effective policy sounds like active inference

i take preliminary steps towards answering these questions in this paper first by examining closely close first by examining closely key formal elements of the framework and the implications they hold for its utility and

second by highlighting potential modes of interpreting the fep in light of an abundant philosophical literature on scientific modeling so quite the um perspectives for the paper and let's see how we're going to get from a to z so here's the roadmap and i had to even reformat the roadmap slide because there were so many subsections but it really helped because it's a paper that doesn't have figures and appropriate enough for a paper called the math is not the territory there isn't any math equations in the paper but it references a lot of quite technical areas so it's kind of like this

fractal overview because you could dive into the thermodynamics or dive into the bayesian mechanics

but this is the overview and it's generalizing

so it starts with an abstract and introduction and then begins with the sort of topic du jour which is the free energy principle this being a philosophy in history of science paper

it's um framing the free energy
principle in terms of its history and
we're going to walk through the history
in a couple of minutes
the history of the formalism of the free
energy principle itself
leads one to a consideration of markov
blankets free energy and generative
models
which um are things that specifically
come into play
when talking about the fep as it's used
and understood today
reinterpreting the fep is sort of what
we're here for today
is to interpret and reinterpret i think
and then the second or the last part of
the paper
ties many of the topics that were
addressed in the first half of the paper
on the left side of the roadmap to more
general questions in the scientific
modeling literature so for example the
difference between normative and process
models the ideas that came out of
mathematical
mathematical psychology related to
neuroscience
and other fields and then ideas about
models that have targets or don't
so a lot of different cool generative
things that are happening with this
paper
and uh it's a lot to cover so let's just
jump
right in so the only hint fristen
himself has given us to go on with
respect to the meta theory or philosophy

of science of the fep
which is what we're here for today is
that he labels it a normative theory or
normative model
contrasting that with what he terms
process models
and then it continues three fundamental
facts follow
three fundamental facts about the fep
follow from its status as a normative
model
so again we have sort of two extremes
types of things in the philosophy of
science
we have the hypotheses those are the
theories they're empirically falsifiable
and then we have these principles
they're the more normative
non-falsifiable almost axiomatic aspects
of science
so three things are going to follow from
the fact that the fep
is a normative model and not a
hypothesis one of them is that the fep
is not falsifiable and does not lend
itself to the direct generation of
predictions or hypotheses
so it's not a hypothesis two
the fep does not cut at the joints of
nature which is to say that it does not
illuminate the boundaries between
natural kinds
and three the fep does not deliver
mechanisms so interesting claims
the general questions that this brings
me to wonder about is
how do we categorize these kinds of
scientific ideas

so for example if someone says well
that's a theory gravity is a theory
evolution is a theory that means
dot dot dot dot so how do we classify
what type of scientific thing ideas
are whether it's a framework hypothesis
uh
idea more broadly just leaving it vague
or whether it's a principle
how do we map out the implications of
different kinds of ideas
and then very importantly how do our
pre-existing assumptions
influence how we judge ideas so we're
not the neutral arbiter of truth
it's more like we have this culturally
prepared field
and then the idea is like the stimulus
or a seed
that flowers or doesn't in a field so
how does the status of our field
influence how we judge ideas and as far
as the cutting at the joints of nature
that's something that really fascinated
me and we're going to go into the joints
of nature
in a couple of minutes so free energy
principle part three in this
paper and remember that these dot zero
videos are kind of overviews
they're just the notes that i take as i
go through reading the paper
i try to make that exoteric for those
who might appreciate a scaffold
but really there's so many fun details
and so many nice points inside of the
paper so if you're curious
or you want to learn more the paper is

the place to go

section 3.1 the history of the formalism

so this is quite different a scientist

might not start by saying

well the thing you need to know about

this theory is here's the history of it

they might say is the thing you need to

know about this theory is this

scientific

attribute abc but this is a more

historical perspective

the questions most frequently and

fervently asked about the free energy

principle are

is it true what is it true of how

do we empirically know that it is true

these questions

i argue rest on a category mistake so

wrong kind of thing you know you ask how

many miles per gallon does this thing

get and someone's like hey this isn't a

car

that's a pizza they presume that the fep

is the sort of thing that makes

assertions about how things are

cuts at natural joints and can be

empirically verified or falsified

so it's the inverse of these things it

turns out that people are interpreting

the fep

as if it's a falsifiable hypothesis that

makes assertions about how things are

and cuts at natural joints

and it turns out that is not the case in

fact

all of those are the opposite why

because the fep

is more like a principle than it is a

hypothesis

pretty cool so again

what are these three points it's the

idea that a hypothesis

does make assertions about how things

are does cut at natural joints

and can be empirically verified or

falsified and it turns out that

mel argues the pep is not that way

let's talk a little bit about nature and

its joints

specifically carving them so just to

quote from the abstract of a 1993 paper

by ali khalidi carving nature at the

joints

descriptive title it's quite an

interesting point which is

this paper discusses a philosophical

issue in taxonomy

at least one philosopher has suggested

that the taxonomic principle

that scientific kinds are disjoint so in

other words

when we're doing taxonomy classification

organizing of scientific

kinds are they disjoint or might they be

overlapping

for example when we're doing a taxonomy

of insects

there's some groups that are disjoint

like ants

and cockroaches are a disjoint group but

ants and insects

are a non-disjoint those are not

disjoint groups because ants are nested

within sight of insects

and an opposing position is defended

here by marshalling examples of

non-disjoint categories which belong to
different coexisting classification
schemes

this denial of the disjointedness
principle can be recast as the claim
that scientific classification is
interest relative

but why would anyone have held that
scientific categories are disjoint in
the first place

it is argued that this assumption is
needed in one attempt to derive
essentialism

this shows why the essentialist and
interest relative approaches to
classification are in conflict

so in other words if we have an
essentialist perspective

on scientific kinds then we can
probably make them disjoint because if
like you know the red cube is red
and the green cube is green then we can
separate them at the joint and the red
will be red and the green will be green
but if the red and the green are like
ink molecules that are swirling around
in solution

then it's more like it's relative to
its interest relative we could say and
in that case there's a

non-disjointness between the redness and
the greenness which makes it a little
bit more difficult

to be reductionist or essentialist and
demands a relativist point of view
and that sounds a little complicated it
is a little interesting

and in 2011 there was a book called

carving nature

at its joints they just uh go with the
same title

because it's the same ideas and in a
review by pd magnus

i thought there was a couple of points
that were just really crisply brought up
so petey magnus wrote to be fair part of
the problem

with this book i believe is with natural
kinds themselves

the phrase natural kind is too often
used as a term of philosophical jargon
with a presumption it has a precise
meaning

even though there are a number of
separate precise meanings it can have
first natural kinds may refer to
categories which will support
inductive inference second natural kinds
may refer to whichever categories ought
to appear in scientific accounts of the
world

third natural kinds may refer to
metaphysically robust categories which
are written in the fundament
of being or which structure the space of
possible worlds

fourth natural kinds may refer to
categories which appear as predicates in
the laws of nature

fifth natural kind may refer to
categories which anchor a special type
of reference

although these five aspects do overlap
somewhat there are difficulties with
each and tensions in between
essays in this volume document some of

these difficulties

so when we're trying to understand this
topic this area reduce our uncertainty
and we want to know about separating
natural kinds it's not quite so clear
what these natural kinds are or what
joints do or don't separate them but
that's what we want to pursue
and so that's why it's fun how could we
go

even deeper down this natural kind
rabbit hole well

we could go to the original text it's
what we're going to do

this is from plato's ephedras in 265.

so socrates writes we can think about
dividing things by classes

where the natural joints are and not
trying to break any part

after the manner of a bad carver so
that's this idea of carving at the
joints

if there are two things and you're not
making a clean separation

then you're going to make this violent
cut but if it's a perfect separation

then it's going to be a clean cut

at the joint as our two discourses just
now assumed

one common principle unreason so in
other words

is it that reason is a principle and
unreason is something disjoint or

are reason and unreason in a continuum
just like the red and the green from the
previous example

so in other words natural join on the
right side can we separate class a and

class b

perfectly or are they going to be a

little bit more intercalated

just as the body which is one is

naturally divisible into two

right and left with parts called by the

same name

so are two discourses conceived of

madness as naturally one principle

within us

and one discourse cutting off the left

hand part

continued to divide this until it found

among its parts a sort of left-handed

love

which it very justly reviled but other

discourse

leading us to the right-hand part of

madness found a love having the same

name as the first

quite interesting but divine which it

held up to view and praised as the

author of our greatest blessings

so are the left and the right two

different natural kinds that we can just

cut down the middle or are they a

continuum on a type of thing

phaedra says very true and then this

last little quote from socrates

even though i know we're quite far

afield now i myself phaedris

am a lover of these processes of

division and bringing together

as aids to speech and thought so in

other words sometimes

just separating things into natural

kinds or continuums is useful for

thought

and if i think any other man is able to
see things that naturally can be
collected into one and divided into many
him i follow after and walk in his
footsteps as if he were a god
and i like this also because it's from
tufts which is the institution of mel
so i just loved grabbing the phaedrus
text

from mel's institution for this
extremely interesting historical dive
let's go back to the paper in order to
understand the fep

however and why it's closer to a
statistical technique

than it is to a falsifiable theory of
biological self-organization it is
important to see that there is a clear
precedent

for leveraging the maths of statistical
mechanics as a method

for bayesian inference so it's like whoa
some people know about philosophy
some about statistical mechanics other
about bayesian inference

but we're going to bring them all
together which means that we're all
going to be learners

here and that's why it's great to
include everyone in the learning journey
because no one is going to read a
paragraph like this and just dust off
their hands and go

yeah i already finished learning about
those topics i'm done learning about
statistics or

i'm done thinking about philosophy those
who are

awake and want to learn and want to find out
other people's thoughts on these questions just come right ahead
section 3.1.1 is called the epistemic turn or the knowledge or the informational term in statistical mechanics
so just a couple of key points because this history section really is quite critical
so this is just in the order of these sections a couple of memes or key themes that i pulled out
so what was happening in the 1900s in statistical physics and statistical mechanics
is that information was getting physicalized
and physics was getting informational so all of a sudden
there was a little bit more of a recognition
that physical things could be equated with information for example if you're talking about the surprise of reaching into an urn
and pulling out different kinds of balls that could be equated
to the surprise that you might have of different temperature
air molecules swirling around in a room
so if
all of the hot molecules went on one side it would be something that would be quite surprising just like if all the blue balls went to one side of the urn
and all the green balls went to the

other side of the urn
so that would be something that you
wouldn't expect and it's through
analogies like that that people started
to bring equivalency
to the informational and the physical
side of physics
the maximum entropy principle emerges as
something like
an occam's razor for info thermodynamics
so again if we're thinking about an urn
with a different colored balls or
a room with a bunch of molecules of air
this maximum entropy
theory is going to basically say well
the air molecules are going to be
maximally spread out
and the distribution of their speeds is
going to have such and such
characteristics
and it's almost like an occam's razor
principle
because it's like saying yes it could be
another way
but this way with the air molecules
equally spread out
it turns out that's the most likely one
because there's an increasing
infinite number of ways that it could be
an increasingly less likely ways
of organizing around the room all in
this corner all in that corner
half in this corner half in that corner
but the maximum entropy way
is spread out and that there's more
details in the paper and of course
there's more to learn about here
um but the 3.1.1 is just to say that

during the 1900s maximum entropy and
this info physical nexus
starts becoming a more important
threat 3.1.2 the mean
field approximation so the mean field we
can think of
is like as you get a field of vectors
you can start to kind of blur your eyes
and look at the mean field approximation
and so one way to think about that is
like the central limit theorem
so if we have a distribution and we're
just pulling out
um with a gaussian error out of this
distribution we're going to approximate
um a draw around the mean value
of that distribution so that's the
central limit theorem
just from 30 000 feet and the mean field
approximation
is kind of like the central limit
theorem because it's basically like
saying
in the mean limit there's going to be
this asymptotic approximation
to collective behavior that might be
comprehensible
so as we get to more and more and more
cars we can start to model them like
they're a flow
of a fluid even though we know that
they're discrete and so you can't do a
flow model with three cars but maybe
with three million cars you could
and this is getting us to this point of
idealized multi-agent systems
so not just draws from a distribution
but potentially

there could be multi-agent systems like
lattices of electrons
or chemical solutions that also have
these mean field approximations
there's then this introduction of the
variational methods
and the variational free energy it's
brought up as a trick
but who's tricking who are the
mathematicians tricking everyone else
are the scientists in on the trick are
the engineers also tricking everyone
um it's a joke but the variational
methods and the variational free energy
are definitely a heuristic and an
extremely useful tool
that help us take something that might
be intractable
and then perform computation on it in a
way that's good enough
free energy which is a concept that
again just like a lot of things in this
paper
have a physical um analog or a physical
tether
earlier in their history like
temperature is related to
the distribution of the air molecule
speed in the room but then temperature
in a machine learning model is something
like how fast
it's trained like a high temperature
model it sort of never settles down it's
always just jumping around
whereas a low temperature model is going
to be trapped in a local region
but it might come to a higher sampling
uh

intervals in that local region
mel in the live chat mathematician eyes
gober
so true and the free energy in machine
learning
came in when machine learning
researchers were asking themselves
how are we going to model multi-scale
systems at the appropriate gain
or bias so we don't have an outrageous
false positive or false negative rate
we also don't have the ability to go and
do an atom level simulation
even if we wanted to we couldn't do
these atom level simulations so how are
we going to coarse grain
and make models that are useful but
don't go into every single possible
detail
and one question or thought i had is
is this variational free energy physical
for molecules so when we talk about
gibbs free energy
and in about in a given reaction a given
chemical reaction the free energy
of that reaction is that a physical
thing or is it energetic or is it
conceptual and then when we're talking
about the free energy of a neural
network is it
physical or not but in this 3.1.3
we get a sort of grand connection
between
ensembl densities so the wood lice on
the pavement you know the ants in the
colony
that density of the ensemble is the
bayesian posterior density so the

updated prior

is the posterior and the posterior is going to be the prior for the next time point but

the updated prior is the posterior and that

is going to be a mean field

approximation and that's going to be governed by this expected free energy in section 3.1.4 the variational bayes or

vb method is brought up and here's where vb comes into play

analytical bayesian methods are basically intractable

and can be implausible to calculate exactly for example if you're doing

bayesian phylogenetics

and you have a bunch of insect genomes

and you're trying to infer

the tree that is the most likely

hypothesis about

common ancestry amongst these insects

to make an exact analytical bayesian

model for every base in the genome

you're going to be calculating a while

however

we can empirically explore probabilistic

graphical models in order to simplify

that

situation and so graphical models are

just relational

models nodes and edges that means a

graph that's graphical so a network is a

graph

doesn't always mean like x y axis you

know quadratic line

graphical just means relational and

we're talking about the relationships
between observations and hidden causes
or two hidden causes
and those are probabilistic
relationships whether it's
probabilistic emission of an observation
given a hidden state
or a probabilistic relationship between
two different hidden states they're both
graphical models
and the idea is that we as investigators
want to fit our model using monte carlo
sampling and or metrics of
distributional divergence
so monte carlo sampling it comes from
gambling
because monte carlo is where a lot of
gambling occurred i believe and
the idea is if you want to ask a
question like how likely am i to get a
deck
uh draw from this deck of cards that has
such and such characteristic
there might be some that are really easy
to calculate like okay the chance of
getting a two well there's four out of
52 cards are two so there's the
probability
but there might be another type of
question you want to ask about that deck
that's a little bit more nuanced
like what's the chance it's this card or
that card but not this one and it's
followed by this
and it has to do with this what you can
do is instead of even trying to specify
out the formal or analytical probability
what you can do is just simulate

thousands or millions of draws
from a simulated deck and then you can
look and say okay
10 out of a million had this
characteristic ergo that's my
empirically derived probability so
that's using monte carlo
or metropolis hasting coupled sampling
to
figure out um the tractable way to do
these bayesian approaches
another way that you can actually
trackably deal with these bayesian
approaches
is this variational bayesian approach
which we're not going to go into too
many details
here but basically unlike the sampling
based
monte carlo bayesian methods the
variational bayes uses
the distributional divergence and it
co-optimizes the accuracy minus the
complexity
just again first pass so that it tries
to hone
into the tightest distribution that
actually covers the density
of what you really want to estimate but
doesn't get too tight it doesn't over
fit and then lastly there's this section
on the innovation in kristen's free
energy minimization so if you're curious
about
what carl has done and the google
scholar is stretching on a little bit
too long
that's a great section to check out so

let's um
pull together and just ask where are we
at
with the free energy principle so what
did Friston do
what is the free energy principle and
how was it discussed
in Mel's paper so the first
is the Fokker-Planck and Mel writes
in the context of the free energy
principle the Fokker-Planck equation
describes the evolution of the
probability density function of the
state of a system
as such it can be thought of as a
trajectory through one
abstract state space which is a
probabilistic representation of some
lower order abstract state space
representing what state a given system
is in
over some definite time window so it's
like
it's a um density evolution function
that's a trajectory
in an abstract probabilistic space
very close to the Fokker-Planck
we can supplement the Helmholtz
decomposition
so let's think about that definite time
window
from the Fokker-Planck pickup here
a three-dimensional vector field that
satisfies the appropriate conditions for
smoothness and decay so smoothness
means that it's not a rough path in
other words you can take a ruler
and just lay it on the hill and get a

derivative and you can
kind of take that ruler and move it over
the hill and get
continuous derivatives everywhere
um if the appropriate conditions for
smoothness and decay are met
then this vector field can be broken
down into solenoidal
which is a curl and a rotational which
is a divergence component
this is known as the helmholtz
decomposition the fact that we can
perform the helmholtz decomposition is
then known as the fundamental theorem of
vector calculus
so in order to have fields of
well-behaved vectors
you need to have this ability to kind of
separate out
and take the partial derivatives in a
few different ways that's the
decomposition
so we're going to be thinking about the
time evolution
of a trajectory in the abstract state
space that's
factor what is that state space going to
be decomposed into
well it's going to be decomposed using
the helmholtz decomposition
into a few different vectors some of
those vectors are going to be solenoidal
they're going to be like curling around
the mountain at the exact same elevation
that's like an iso contour
and then there's also going to be a
non-rotational divergent
component one which goes uphill and one

which goes downhill
now we're going to be doing that on
non-equilibrium steady-state densities
this is related to this idea of the
markov or
post-markov blanket i don't even know
what to call it
um mel and anyone else let's clarify
this when we're talking in our group
discussion a fristen blanket or a yellow
blanket what are we going to call them
but when it comes down to it
when we want to be maintaining this info
thermodynamic non-equilibrium steady
state
it's required to have a relatively
better because you don't need to have
the perfect one you just need to have a
relatively better model
than other agents like you you need to
have a relatively better
generative model of what of action
not necessarily just of accurate state
estimation
so the bacteria isn't just doing
estimate oh oh boy we're running out of
sugar
i wonder if there's uh sugar somewhere
else it's actually doing
inference on its behavior on policy its
actions
so that it can maintain that info
thermodynamic
barrier mel writes markov membrane
that's my contribution
if only we could all make such a great
contribution
this need for a maintenance of an info

thermo non-equilibrium steady state and
therefore a generative model of hidden
causes the world entailing policy
selection paths
leads to this imperative to be
self-evidencing
or an information forager that takes
this multi-scale cybernetic
goal seeking good regulator requisite
diversity characteristics
so in other words if you want to be
staying alive maintaining that
asymmetry where you have more
organization or more information that's
that info thermo
if you want to be maintaining that
competitive strategic asymmetry
then you're going to be needing to have
the ability to engage in effective
policy
effective policy is only going to be
happening if you have an effective
generative model of the latent causes in
the world
and that means that if you're going to
be a cybernetic agent
that's able to have a generative model
about the latent causes of the world and
respond appropriately
you're going to be needing to be an
information forager
that has these good regulator and
requisite diversity
characteristics that is what's subsumed
under the free energy principle
now what are the corollaries so the
corollary hypotheses
or theories are those that are going to

be making specific
testable hypotheses or multiple and a
couple of them that we've brought up
today we didn't write the ones that
according to the alias interview didn't
fare so well like gibbs
particle filtering or gibb sampling but
we can write up here active inference
predictive processing
bayesian brain and what else so if you
have any
other ideas for what is a corollary to
the free energy principle let's
hear about it that'd be fun and each of
these three
make specific predictions about kinds of
observable things you might observe or
the implications of different kinds of
perturbations into the system
so these are a lot more axiomatic and
background more foundational
whereas up to the top we see in the
corollaries these are
more like theories that make specific
testable hypotheses
for example active inference could be
falsified if somebody shows that
external states are able to alter
internal states without
going through sensory states for example
so that's the free energy principle
in this next section which is 4.1 we
will turn to the implementational
details of the fep
which is where it will be demonstrated
that the formalism does not latch onto
any features of the real world systems
of interest

so we sort of built up how the fep
draws in historical context from
info and thermodynamics and then draws
in this cybernetic thread
to result in a panoply of these testable
hypotheses
so let's talk about section 4.1
but before we jump in just general
questions and this is because
it's easier for me to come up with
questions than answers is
what kinds of models do latch on to
features of real world
system of interest and what is
of interest what is of interest and how
is it related to what is useful
for basic and applied researcher or for
the philosopher of science
so for example if your friend says hey
you might be interested in this
well for the applied researcher or for
the engineer
maybe of interest would be this is
useful hey you might want to learn about
this software package it might be of
interest to you because you might use it
effectively
to a basic researcher somebody who's
working more in the theoretical or the
abstract areas
of interest might be like you'll find
some pattern in here
that's very interesting you might have
some thought cascades that are exciting
and um so it's a different interest and
the philosopher of science
what is of interest for them i don't
know

let's trace a little bit about this
markov blanket
question and um the uh
markov blanket we're gonna return to it
a lot more in
later on in the year when we are
in april on this uh emperor's new markov
blankets
and in a few other cases we're gonna be
talking about marco blankets
but again we're always learning about
these topics so not too surprising we're
going to come back to them
multiple times so this is what
mel wrote i i love in the live chat
people are coming up with all kinds of
funny
markov related terms markov train chains
by marco
just yes chains of metaphors metaphors
for chains
why not we come to the conclusion that
within any non-trivial network any node
ought to have a multitude of markov
blankets
fristen's employment of the markov
blanket construct differs from pearl's
original 1988
construction in the subdivision of the
blanket into sensory states
nodes whose influence is directed
towards the blanket of nodes acts
and active states nodes influenced by x
okay
so here's 1988's book by judea pearl
probabilistic reasoning in intelligent
systems
network of plausible inference now this

idea of bayesian computation on
graphical models or bayes nets
is discussed in this book and elsewhere
the markov property
though there's a couple things that have
markov's name attached to it and i wish
there was often a clearer way than just
invoking
some person's name but we'll just use it
the markov property
in probability theory and statistics
refers
to the memory list property of a
stochastic process
the term strong markov property is
similar to the markov property except
that the meaning of present
is the firm defined in terms of a random
variable known as the stopping time
and so the term markov assumption is
used to describe a model where the
markov property is assumed to hold such
a hidden markov systems
hidden markov model so for markovian
systems
we can think about a blanket that is
insulating a node
x now as mel pointed out in pearl's
original construction
it was in a graphical general abstract
context
and the node x was insulated from
the broader network by a blanket set of
nodes
fristen was going to do something a
little different
which is the nodes that are incoming
into the node of

interest so here μ is the internal state that's like the x node from the bayesian notation the nodes that influence the node of interest are the sensory states that's like the incoming states but again it's it's whichever perspective is inside or outside so we're speaking from the point of view relationally of this internal state the sensory states are coming in and the action states are the ones that are downstream and so notice that the sensory states actually include states that don't directly impinge upon internal states of interest but actually relate just directly to active states and so that's this diagram we've seen as the the fristen markov blanket as a statistical boundary with the parents children and the parents of the children so it's kind of like your family for the person of interest includes this person who maybe they're not directly related because it could have it doesn't have to be every arrow on here but they're apparent of their child so it's important now again it would be awesome to have people who know better and have thought about it more to give their perspective but i just tried to make a little bit like a continuum of markov blanket ideas

so on the left side we have the actual
markov

our colleague from russia over here
and maybe i'm kind of straw-personing
these arguments i'm just trying to
throwing them up there trying to
understand

where we're coming from and where we're
going so markov

stated it using mathematics pencil and
paper

and this side of the spectrum is held
down by

the abstract the general and no need to
appeal

to computational bases pearl

is somebody who had a computer not
computers like we have today but
good ones nonetheless i believe pearl's
framework

is based upon wanting to integrate
empirical data in a way that mathematics
isn't so that's one of the distinctions
between statistics and mathematics
because it's dealing with statistics and
um

uh real data sets potentially weaker
assumptions

might be in play so this is sort of
somewhere in the midpoint

and fristen is way on the other side of
the overton markov window

and this is where we're getting the
integration of ideas like

the node consists of sense and action
states it's not just

insulating states it's actually there's
this temporal nature

where things that happen before
are like sense states coming in and
things that happen afterwards are like
actions heading out
and that's like a bayesian directed
graph that has a temporal component
we also bring in these attributes like
this cybernetic imperative
to maintain the non-equilibrium steady
state the idea of a generative model
of non-local dependencies or latent
causes in the world
cybernetic model of the environment the
good regulator or the requisite
diversity ideas
and also some of these other threads
that might not be
downstream of some qed proof from markov
but nonetheless are important for
understanding of markov blankets in the
free energy principle
like the inactive embodied and cultured
embedded
perspective so
can the markov blanket idea have
interpretable parameters so do
any kind of blanket capture what we
think they're capturing or what
should we think they're capturing is
this implementation tractable
will it be useful for different kinds of
systems
is there clear philosophical commitments
or not
and is narrative thermocybernetic a
metaphor or not just you know i'm
curious about that
um a few more quotes from the paper and

thanks everybody who's watching and
posting in the live chat it's pretty
fun and a little hard to be responding
to everything but it's fun
times mel writes
this is how the markov blanket construct
is operationalized
the upshot is that the power to select
systematic boundaries rests at least in
part on the researcher's intuition which
i would also say is true for regression
coefficients
principal component analyses t-sne
dimensions okay that's just
what it's like being a scientist we
didn't sign up for this because we
wanted
objective non-personalized
recommendations we wanted to be part
of that process if the markov blanket
formalism
were to independently track natural
joints we would need to equip it at
least
at the outset with some threshold value
which would determine precisely the
degree of conditional independencies
that would
count across the board for the
possession of a markov blanket
so that's like when you do principal
component analysis on a real matrix of
data
let's say principal component 1 explains
40 percent of the variance and then the
next one's 30 and the next one's 10
and you get this long tail of hundreds
and hundreds of principal components

that explain
less and less and less variation there's
no clean line
you can't just say well if it's less
than one percent of the variance
then it's not an important principle
component there isn't
a measure or a value like that the
threshold is always going to be
in the details it's always going to be
how it's enacted by the scientist
if such a threshold existed ask yourself
if it could
and were baked into the markov blanket
formalism
then we could consider markov blankets
to be in some sense real features of
real world systems
so again like p values you can say i
mean come on
p value of 0.00
come on let's just say that they're
different but no you you can't do that
because there's no
natural kind with p values it's just a
continuum
we might discover them measure them and
count them
we might meaningfully ask whether some
existing system does or does not possess
a markov blanket
so again let's just say there's two tree
species and one is ten plus or minus one
foot and the other one is three plus or
minus one foot there's no overlap
and you do the t-test and the p-value is
just ridiculously low
and you say come on can't i just say

it's a natural kind that they're
naturally different
and so the answer is no so this is a
similar realization in the context of
the empirical detection of markov
blankets potentially from empirical
systems
if we can conceive of something as a
discrete thing as a coherent system
then it is possible to formally
represent it as possessing a markov
blanket
there are no markov blankets to be
discovered in nature and they are not in
the business of illuminating natural
joints
so it's like there's no p-values in
nature there's no structural equations
out there in nature
we're not discovering the structural
equation model we generate it
in response to observables so that's why
the ftp
is not in the business of illuminating
natural joints
finally the thermodynamic entropy and
the shannon entropy
are only equivalent under the
generalized boltzmann distribution
which it has been argued applies only at
thermal equilibrium
thus in general information entropy and
physical entropy are distinct
living systems are by definition far
from equilibrium systems
thus information entropy and
thermodynamic entropy
do not converge in the regimes of

interest to us

so just a couple general questions on

this part

what are the formal axioms and the logic

or ontology of the free energy principle

this is what mel has taken such an

awesome

effort and a stab at with this paper is

what are the commitments and entailments

of fep

what did we have to do to get here where

are we

and where are we going to go

philosophically

which of the model assumptions of active

inference or

fap are flexible so where would we find

that the model

is still performant on data sets that

might violate some expectation of the
model

so for example a t-test might have

the assumption that there's equal

variances between the two groups

you could do an unequal variances t-test

it's a different kind of test but for an

equal variances t-test it assumes that

the variances are equal

but they don't have to be equal to the

highest possible degree

it turns out you can kind of fudge that

one a little bit or it might still give

you directionally accurate findings

but with potentially a reduced or an

inflated p-value

so in other words what assumptions are

we bringing to the table

and what are the entailments and then

which one of these things bringing to
the table are a little bit flexible
what are living systems doing and how do
we think about living systems this is
related to that life
mind continuity it's related to
schrodinger's question
what is life from 1944 it's related to
complexity theory it's related to so
many discussions that we're having about
agency autonomy individuality
personality multi-scale systems there's
so much
happening here and what other
questions do you have because the
questions that
all of the participants bring to the
table are often the most exciting
ones it might seem then
that the fep could perhaps be thrown at
real world systems and the degree of
biological or cognitive complexity
can be read off of features of the
system's generative model
the fep however does not dictate what we
should write down
for a given systems generative model and
once we have written down the generative
model of the system under investigation
we have switched from the purview of the
fep to that of a process theory
so again let's think about it with a
t-test the t-test
by itself doesn't dictate which exact
statistical test we should do for a
given system
so the t-test framework doesn't say yeah
this is what you measure

about ant colonies the t-test says given what you give it it's going to give you a p-value and so that's something that the fep is analysis analogously going to be providing us with so as soon as we start asking about specific deployments of the fep in situ in an actual case then we're talking about falsifiable process theories so it's almost like linear regression theory is in the background that's abstract and then you can ask in specific situations whether you're using um a falsifiable model using a linear regression on a real data set but that's not at the level of talking about what linear aggression is which might be an in principle discussion this i thought was an interesting part as well a single hydrogen atom at rest in a vacuum is not meaningfully interpreted as performing approximate bayesian inference over its environmental states a time series of mere abiotic self-organizing systems say a whirlpool or a candle flame would show it throwing off its markov blanket and establishing a new one at every time step critically this does not entail the fep cannot in principle apply to such systems it means

only that we have no reason to apply it
to such systems
we have nothing to gain epistemically
from applying it to such
systems so that would be quite
interesting to
hear mel's perspective on and everyone
else as well
what is different about systems that
seem to just kind of
dissipate and don't have this
autopileisis characteristic
like a candle burning or a landslide or
a lightning strike
they're energy dissipating events but
they're a little different than living
systems and so it makes me think of
le vocier working with oxygen and
thinking about how well
humans are taking in oxygen and
breathing out CO_2
candles are taking in oxygen so to speak
releasing heat
breathing out CO_2 kind of the reverse
plant
reaction so are they similar are they
different what's what's happening there
just a couple more quotes this is really
fun though
we found that the framework accommodated
certain things very well
this is mel in review at the latter part
of the paper the markov blanket
does a very nice job of representing
systemic boundaries
the temporal depth of the trajectories
being solved by the latent generative
models postulated under the fep

is a very elegant and informative representation of something like cognitive complexity and the free energy parameter itself maps onto a system's attunement with its environmental context its cohesion and internal consistency among other things we also found however that the fep does not pick out or discover these aspects of natural systems even in silico but only provides a useful model of them so this is really awesome because it just is such a positive example of the yes and mindset that we want to be fostering with active inference and fep we're drawing from all these different areas areas where we know we're going to be learning for our whole life about information theory and cognitive complexity and machine learning we're drawing hopefully together different ideas leaving the door open for those who can contribute in ways that we don't even know or expect building together and then also just clarifying the fep is not doing everything it's not picking out or discovering aspects of natural systems even in a simulation however it may yet be something quite useful for us to deploy and this is just a few more points from

the modeling section on this paper
yep it's just it's a saturday it's a fun
paper

we're looking forward to 14 and so mel
writes

this comes in two forms first the
literature on scientific modeling
has increasingly come to acknowledge the
status of non-representational
and non-target directed models and
second

even at the presumption that all models
represent target systems

the utility of a model is generally
understood to stem from idealizing
black boxing or core screening away from
essential

inessential or distracting details of a
target system

in other words deliberate deliberate
misrepresentations

or omissions make a good model not
fidelity

so very interesting it's a semantic
compression and this question of lossy
or leaky compression

and in the context of scientific
modeling

it's related to whether we're talking
about representational models or non
or target directed or non-target
directed

so first point here how are models
scientific models or other kinds of
models that's the demarcation problem
related to semantic in compression how
can we think about

semantic compression as encryption

and it could be encryption that's
reversible or not
how is semantic compression related to
narrative how are narratives
semantic compression events how are
narratives the stories that we tell
able to compress vast swaths of syntax
or even other semantics how is
coarse graining or removing the
inessential or distracting details of a
target system
discarding irrelevancies as bucky would
say how is this coarse graining
related to the causal entropy of a
system or maybe even its slow
synergetic modes of operation
good models live on this complexity
accuracy trade-off
in the philosophy of science space so
you want your model to be something more
than nothing
if it's not reducing people's
uncertainty at all if it's not helping
them get more accurate about something
it's nothing but then there are model
complexities that can outweigh
how accurate um or how much it assists
in agents accuracy
so if you have to carry around a hundred
things in your head and then it only
helps you a little bit
maybe you could have had another thing
in your head that would have reduced
your uncertainty in a better way
so it's just so cool that we see this
complexity accuracy trade-off
manifested in the actual formalisms of
the fep

but at the same time qualitatively it's
long been understood in the philosophy
of science
that models have to ride this edge
between being too complex
and being accurate can we separate
these boxes at the joints for use and
truth and so i forget which discussion
it was on or who said it
but it was um maybe an in eight with uh
blue or with alec
how when we want transparent machine
learning models
this idea we're going to look inside the
black box potentially might be a little
bit misleading because you're going to
look inside and there's going to be
millions of parameters
so just looking inside of the box may
not get
semantically somebody to where they want
to be however
could we cut up that box into specific
structures of smaller boxes
and then we could look at how variables
or information is passed
between different boxes to be able to
have a sense
of what is almost like accountability or
transparency in machine learning models
so just to give one example there let's
just say
somebody has a data set with movement
data and
pictures of people and there's a neural
network that's predicting whether
they're going to be committing a crime
and somebody says hey i think this is

being used in a non-equitable way i want
to have increased transparency
i want to be able to see inside of the
model and someone goes sure
here it is look at all the parameters
yourself but again they're not
interpretable
they're parameters inside of some neural
network model so
would there be a way where we could
separate the model
into these specifically related boxes
that are doing message passing
passing information back and forth in a
way where we can see
oh yeah you know you told me that
socioeconomic factors
don't play into this part of the model
and now i can actually
look at that and see yes these variables
are getting
concluded into a summary variable and
that's getting combined with
socioeconomic
status in this way so it's not playing a
role at this stage of the model
it could be confounded because it's
confounded in the real world but
by breaking up the black box into
smaller boxes that are a little bit more
graspable a little bit more
comprehensible
we actually get a level of understanding
that isn't just peering inside of the
model because again
do we really understand a linear
aggression do we really understand
addition

i mean you can really go down the rabbit hole do we understand a sign or a cosine lots of fun stuff happening today all right we got a couple more slides another quote from mel in the latter half of the paper the organization and dynamics of the living organism the functional architecture of the brain and the structure of human social systems these are the most complex systems known to exist and they're nested inside of each other the systems that study these systems oh sorry the sciences that study these systems are comparatively very young and may never reach the maturity of the sciences oriented towards far more simple systems the life cognitive neuro and formal social sciences are still in many respects at the stage of trying to get a methodological foot in the door highly idealized models such as normative or optimality optimality models assist us in getting traction on otherwise intractable phenomena so we saw that a little bit with the scripts paper and the variational approach to scripts because that was a social science that was methodologically getting its foot in the door by borrowing from the legacy of the kinds of approaches we're discussing here loose 1995 introduces four distinctions

to the mathematical modeling
of cognitive processes so this is about
the cognitive psychology mathematical
psychology
science modeling literature process
models are contrasted to
phenomenological models
normative models differentiated from
descriptive models
dynamic models compared to static models
and a distinction is drawn between noise
and structure
so that's another breakdown of models of
kinds of scientific ideas
not fristen's breakdown but loose's
breakdown and that also
helps us understand a few more
spectrums that exist in the space of the
possible models
another quote from the modeling section
under a normative model
which is again the kind that we are
suspecting fep
is it is presumed that reasoning should
accord with formal logic
induction and beliefs with eg bayesian
dictates for inference and credence
belief or decision making with the
results of optimizing a utility function
a descriptive model or phenomenological
model in contrast
represents the cognitive process of
making choices
reasoning through sub-problems and
drawing inference as it is observed to
happen
messy sub-optimal and irrational though
it may be

what then is the significance of taking
the fep to be a normative model in
loose's sense
fristen and colleagues stress that the
fep as a process theory is not
falsifiable
i'm not sure if this is intended
it will not be possible to articulate a
version of the fep that can be held up
against some real world process in such
a way as to undermine or legitimate the
model
because the fep is the principle that
specific process theories
are drawn from the fep will not directly
generate predictions
tests or hypotheses well i would wonder
what if it brings the group
together and then they come up with
predictions tests and hypotheses could
you say that the
idea was the difference making cause
just a thought
another idea on this is the normativity
here is presumed to be a logical
normativity in other words that the
reasoning should accord with formal
logic induction and belief
but what if the normativity were
cybernetic for example what if the
normative claim
were not that a can't equal not a
so that's sort of one of the claims of
let's just say
normal boolean logic but one of the
claims is
the normative claim is uh might is right
for example so this describes cybernetic

systems

which might not be following formal

logic or pianos axioms or

any other inductive strategy that's

consistent with truth

they might be following normatively a

power centric approach

now if that diverges too strong from

what's actually there

they're not going to exist in the niche

but still that doesn't mean that the

normativity

has to be derivative from this reality

it means that it could be a normativity

that arises from natural selection

and also marco thanks for that comment

wrote i want to flag this part in my

opinion fep

is a unique normative principle that

doesn't prematurely assume

sound reasoning so that will be great to

unpack

your understanding there and um

just a few more quotes in the section

that follows

the philosophical literature on adjacent

sorts of models will be reviewed in the

hopes that this will lend a sense of how

it is that a highly abstracted

unfalsifiable formal model such as the

fep can have scientific utility

ultimately of course whether the fep

turns out to be useful in this way will

be an empirical matter

in both senses great puns here it's like

if it's really true that the fep is

useful then we'll find out because it

will be useful

and how could it be that a highly
abstracted
non-falsifiable formal model could be
useful
that's what we're finding out we're
learning by doing that's what we're
enacting
the fep is thus an umbrella framework
out of which predictive processing
predictive coding
and aversion of active inference its
process models
fall with decreasing abstraction and
increasing granularity
awesome sentence as a normative model
the fep is intended
to aid in the generation of process
models and to furnish constraints on
viable process models
the fep itself is however not beholden
to empirical data
its virtues are not in its very
semi-limited
very similitude reminds me of the
silmarillion but maybe there's a better
way to say that word it means
truthfulness though
so this is just really awesome it's kind
of like by
allowing reward learning to evolve into
active inference
we saw how preference becomes baked into
the model so we don't separate from our
values and preferences by focusing on
policy
actually focusing on policy is what
allows us to be clear
and define our values and so similarly

by
going to this highly abstracted
unfalsifiable formal model such as the
fep
we're able to clarify where scientific
utility
is going to enter the picture where in
specific situations
task specific modeling or setting
specific modeling
and it's going to be developing
falsifiable hypotheses
but the fep itself there's no data set
that comes along and knocks out the fep
so it's kind of at the level of
evolution rather than at the level of
these process theories all right just
the last couple few i feel like i said
that a few times
this is the field site where i did my
ant research in grad school in arizona
but i like this because uh we'll see
in a in a little bit weisberg
has likewise laid out a useful taxonomy
for scientific models
he differentiates concrete mathematical
and computational models
cutting across this tripartite
distinction is a division between what
he calls
target directed modeling and modeling
without a specific target
in this latter category of target list
modeling we find generalized modeling
minimal modeling hypothetical modeling
and target list modeling
a target directed model is one built for
the purpose of predicting controlling

and explaining the behavior of a
specific system under specific
conditions

for example just what an example we
might have data pertaining to the
foraging behavior

of a particular species of ants across
a number of colonies observed in the
amazon rainforest and we might construct
a model for the purpose of discovering
something about the
ethology and ecology of this particular
ant species

then again we might want to think about
foraging behavior much more broadly and
abstractly

divorced from the minutia of particular
minute particulars

william blake populations of particular
species in particular ecological
settings

considering the problems posed by
foraging in apache habitat as faced by
organisms in abstract terms

we might perceive a resonance between
foraging and economic models that deal
with optimizing decisions in scenarios
in which we face diminishing returns we
might come up with a heavily idealized
organism and context

generic model of foraging the latter
model

the latter approach is what weisberg
dubs modeling a generalized target
so at this field site for over 30 years
my phd advisor professor deborah gordon
was observing

exactly the specifics here the minute

particulars

the very specific behavior of ant

colonies at this field site

now there was other ants other species

at the field site

there was the same species in the next

valley over but we were really specific

about this specific field site these

exact colonies we'd have stones next to

the colony oh

here's colony 6 10 and so

that's specific at the same time it was

through observing the specifics

that one is able to abstract and

generalize for example

um professor gordon's papers in 2014

on the ecology of collective behavior

which draws links between the algorithms

from computer science

and behavior across ants as well as

swarm behavior in other systems

so it's like by going into the specifics

we can go into the abstraction

and we can make it even more clear it's

kind of like an electrical charge

separation that we're polarizing it

because it's

clarifying and distilling the hypothesis

and the specific ants on the ground as

it were

from these more general ideas about

ants foraging the live chat goes wild

with the foraging behavior in ants

models of this sort can also serve as an

abstract or analogical touchstone

or entry point into an unexplored domain

so this is really exciting models as

exploration

this will serve as the basis for more
targeted more fine-grained modeling work
or empirical investigation later on
models in this
genre also aid in the process of theory
construction
by allowing exploration of the nature
and interrelation
of conceptual objects and by inspiring
theorists to draw connections and
ask questions that would otherwise have
gone undrawn or unasked
such models offer leverage where other
scientific methods stop short
systems and dynamics far too complex or
too new to be treated under standard
approaches
so with the ant example you're out there
in the field you're watching the ants
forage
and you start to abstract and generalize
and think about
optimal foraging or think about general
foraging processes
but now we're stepping out even a level
beyond
the idealized process in nature are
these abstract
entry points into unexplored domains for
example
the game of life which is a cellular
automata game
it was like put out there as a toy as a
game
yes everyone is losing their mind in the
chat mel
i hope that you can recover by the time
we have this chat next tuesday

the game of life was just like saying
hey what would happen if we had a grid
with a certain rule set
and that ended up going so much
in detail people making gliders and
turing complete
games of life and stephen wolfram and
all the research downstream of this
so these models it's just really nicely
placed
you have the targeted versus the target
list models
so the target models you know ants on
the ground in arizona we're doing linear
aggressions
specific falsifiable predictions and
hypotheses
about this ant species on the ground
we're using that
to abstract more general questions about
foraging like
hey i wonder if there's a patchy
resource
then the ant species will probably try
to recruit its nest mates to take
advantage of a
patchy resource that might not be around
but if it's a distributed resource
that's not patchy
then maybe the ants will forage without
pheromone trails
we're going a level even above that to
this idea of like
generators for ideas
last quote from the paper
figuring out what the fep is and what
use it holds for scientists
is a worthwhile project in and of itself

how we come down on the matter and what
route we take in getting there
and who we include and how we
communicate about it all these things
and more
will have important implications for the
philosophy of biology
philosophy of cognitive science and the
philosophy of science in general
the dialogue unfolding in the literature
on fvp raises important questions
about the relationship between science
and philosophy of science
between theory model and data and about
the scientific
method and the aims of science even
perhaps most especially
about what counts a science in the first
place taking us back to that first
sentence
in the introduction that with respect to
the demarcation
uh problem mel is going to defend the
claim that the fep
is scientific and just by reading this
paper
and thinking through some of these
topics i feel like i
learned a lot about science and about
scientific modeling
so closing notes this is a
project in progress that will include
many backgrounds and perspectives
so let's have a participatory and an
accessible
conversation about active inference
sounds like it'd be a ton of fun
because fep and active inference is

bringing us all
to fundamental conversations from the
philosophy of science
within philosophy of science but also
about how science is related
to the philosophy of science it's also
bringing up debates that are related to
other fields of philosophy
in activism consciousness
computationalism
semiotics autopoiesis communication
meaning evolution time space
all of these fun general topics are
being brought up
through the work that hopefully we're
all participating
in so this was an awesome paper and
i'm really looking forward to the next
two weeks of discussion
we can just throw up our general
questions
what would we enable for ourself
or for our teams if we had a good
understanding if we understood
what the fep was if we understood that
the as marco just put in chat the math
is not the territory but the philosophy
is the landmark
if we come to understand that what will
we enable
what are the unique predictions and
implications of different understandings
that we could have
what are the next steps for free energy
principle and active inference as
research communities
what are the goals of this research what
is the goal

of a philosopher of science who might be
curious about this topic
and perhaps most importantly what are
you still
curious about if you're listening to
this video because
this is a super fun topic it's so
um excellent to have appreciative
co-authors
mel thanks so much dan this was
fantastic well the paper was fantastic
and
really a learning catalyst for i know
multiple of us
so we're just starting off the two-week
uh
period of this paper so everyone
thanks for participating we're gonna
provide follow-up forms for the live
participant
we request feedback suggestions and
questions as always
and everyone just uh stay in touch
it's been a nice stream and a nice
conversation
we're gonna be having let's just get the
calendar up again
january 19th and 26th 2021
we're going to be talking about this
paper at 7 to 9 a.m pacific
mel is going to be joining us for the
stream and so
check out this rb.gy/kvn
pyc link and uh yeah
this is going to be an awesome
discussion so any background
any familiarity you have with the topics
you're

more than welcome to join our group
discussions
and i hope to see you all soon so
have a good day and peace